

Mansi Mantri

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PROFESSIONAL SUMMARY

Software Engineer with 2+ years of internship experience designing and deploying distributed systems, RESTful APIs, and backend services using Python, Java, Node.js, and AWS. Strong foundation in data structures and algorithms, with hands-on exposure to machine learning frameworks and LLMs for solving complex technical challenges.

TECHNICAL SKILLS

Programming Languages: Python, Java, JavaScript (ES6+), TypeScript, SQL, Kotlin

Computer Science Fundamentals: Data Structures, Algorithms, Debugging, Code Reviews

Machine Learning & LLMs: LLMs (DeepSeek, CodeT5+, Llama 3), Hugging Face, RAG Pipelines, AWS Bedrock

Distributed Systems & Cloud: AWS (S3, Lambda, Step Functions, IAM), Docker, GitHub Actions, CI/CD

Backend: Node.js, Express.js, FastAPI, RESTful API Design, Authentication & Authorization

Databases: PostgreSQL, MongoDB, MySQL, Supabase, Firebase, Prisma ORM

Tools: Git, Linux, Agile/Scrum, Postman

EXPERIENCE

VODA.ai | *Data Analyst Intern*

Jun 2026 - Aug 2026

- Queried and maintained SQL relational databases, performing large-scale cleanup, deduplication, and standardization across 10K+ records to support accurate data collection and reporting.
- Built automated reporting dashboards using Power BI and HubSpot, improving data visibility and reducing manual reporting effort by 50%.
- Developed data-driven enrichment scripts and API-based hooks to automate scoring workflows, translating business requirements into functional reporting tools.

Dottir Labs | *Software Engineer Intern*

Feb 2026 - Apr 2026

- Studied the existing system and redesigned the legacy Tkinter desktop app into a modern web application using React.js, FastAPI, and Julia (for data processing), with Plotly.js dashboards for spectral data visualization (spectra, normalization graphs, calibration curves, peak markers).
- Implemented core features including spectral data processing, peak detection with manual override, calibration curve generation, and offline/online connectivity with an offline upload queue and retry logic, plus Google Drive and local storage integration for CSV/JSON data.
- Conducted end-to-end system testing before pilot deployment, identifying and resolving 15–20 error-handling issues - including battery safety checks, hardware/laser testing, and crash recovery - directly enabling the system's successful advancement to pilot phase.
- Partnered with Design and QA engineers to diagnose production issues, review code, and iterate rapidly on fixes.

Park Pulze | *Backend Engineer Intern*

Sep 2025 - Dec 2025

- Developed and optimized scalable backend systems handling high-volume data (2K+ users), improving system throughput by 35%.
- Designed secure RESTful APIs with authentication and access control, improving data synchronization performance by 40%.
- Debugged and enhanced system reliability through testing and code reviews, reducing issue resolution time by 25%.

Crossing | *Software Engineer Intern*

Sep 2025 - Dec 2025

- Built and supported distributed AWS pipelines (S3, Lambda, Step Functions) integrating machine learning preprocessing, delivering high-quality production data processing at 200+ records/week scale.
- Developed backend systems with asynchronous processing and LLM-based validation checks (AWS Bedrock), debugging complex data workflows and reducing manual review effort by 60%.
- Built a RAG (Retrieval-Augmented Generation) pipeline for automated document extraction and filing assistance for attorneys, reducing a 2–3 hour manual filing process down to 5 minutes, while collaborating cross-functionally through structured data validation and AI-assisted debugging workflows.

Atlas Copco | *Full Stack Developer*

Jan 2024 - Jun 2024

- Architected and shipped a real-time microcontroller reservation platform (React/Next.js, Node.js/Express, PostgreSQL) using WebSockets for real-time locking, enabling conflict-free, login-authenticated scheduling - built to scale for adoption by teams across multiple countries.
- Engineered a fully containerized deployment pipeline with Docker, enabling one-command, environment-agnostic deployment; documented all APIs via Swagger and delivered a polished UI with Tailwind CSS for maintainability and adoption.containerised
- Drove system reliability and performance through query and API optimization, resolving 30+ production issues and cutting latency by 20%, while partnering with a cross-functional team of 10–15 engineers through daily scrums to translate feedback into iterative improvements.

Applied Technology Solutions, Inc. | *Software Engineer (Research and Development)*

Jan 2023 - Dec 2023

- Resolved UI/UX issues on client-facing portals and streamlined API calls, reducing reported interface bugs through targeted redesign contributions, while optimizing legacy web platforms for a 20% increase in system performance and user engagement.
- Built and tested backend services using Java and Spring Boot, using Postman for API testing and validation, and managed SQL database migrations by modernizing legacy systems with newer frameworks - reducing data processing time by 30%.
- Implemented NTP-based clock synchronization for accurate system timekeeping, and developed Python automation scripts and Linux-based testing/validation workflows, reducing deployment time by 40% and increasing test coverage and team efficiency by 30%.

PROJECTS

Careerlens AI | [Careerlens AI](#)

2025

- Architected a full-stack AI-powered SaaS application with secure JWT-based REST APIs, PDF parsing, and Google Gemini integration for automated ATS scoring and skill gap detection.
- Designed multi-feature backend with session management, Prisma ORM-based PostgreSQL storage, and AI-driven workflow pipeline, reducing job outreach effort by 80%.

Vizbotz – Data Visualizer | [Vizbotz – Data Visualizer](#)

- Built an internal tool automating CSV data ingestion with customizable column selection, enabling data analysis and reducing manual effort by 40%.
- Integrated AI to auto-generate analytical summaries from querying raw datasets, accelerating data-driven trend identification by 30%.

GompeiMarket – WPI Campus Marketplace | [GompeiMarket – WPI Campus Marketplace](#)

- Architected a full-featured Android marketplace app using MVVM + Repository pattern with Dagger Hilt DI, featuring secure @wpi.edu Firebase Authentication, real-time Firestore listings, and integrated buyer-seller messaging.

- Implemented AI-powered listing creation with Gemini API auto-generating item descriptions and ML Kit Image Labeling for automatic categorization, with Google Maps SDK displaying verified safe meetup zones on campus.

Stock Sensei – Intelligent Stock Market Forecasting Tool | [Stock Sensei – Intelligent Stock Market Forecasting Tool](#)

- Conducted data preprocessing, feature engineering, and forecasting (ARIMA, Prophet) to design predictive ML models for market trend prediction with 95% accuracy, ensuring scalability, reproducibility, and reliable model performance.
- Integrated PostgreSQL for storage and retrieval of 500+ financial records, and designed interactive BI-style dashboards using Plotly with Tableau-like visualizations for real-time trend analysis, enabling data-driven investment decisions.

BigDocBot – AI Documentation Tool | [BigDocBot – AI Documentation Tool](#)

- Built a Python-based code summarization ETL pipeline using LangChain and transformer models (CodeBERT, CodeT5) to extract function-level metrics across large codebases
- Constructed AST-based parser to generate line-by-line LLM explanations and token-based heatmaps.

Arena Vibes – Ticket Booking System | [Arena Vibes – Ticket Booking System](#)

- Built scalable RESTful APIs with Node.js and PostgreSQL supporting real-time booking workflows, optimizing 30+ SQL queries and reducing response latency by 20%.
- Developed end-to-end full-stack architecture with TypeScript enforcement, debugging complex logic to reduce booking errors by 90% and improve reliability by 40%.

Expense Tracker – Personal Finance App | [Expense Tracker – Personal Finance App](#)

- Developed a full-stack personal finance platform using React and REST APIs, enabling users to track and visualize expenses through interactive dashboards built with Chart.js.
- Improved application responsiveness by optimizing API interactions and frontend state updates, reducing manual expense tracking effort by 60%.

SecondBrain AI – Multimodal Study Assistant | [SecondBrain AI – Multimodal Study Assistant](#)

2025

- Built a fully local multimodal AI assistant with zero cloud dependency, using Python for end-to-end pipeline design.
- Designed an automated document processing pipeline with pypdfium2, ffmpeg, and PIL, enabling reliable local inference via Ollama.

EDUCATION

Worcester Polytechnic Institute	MS, Computer Science	Aug 2024 - May 2026
MGM's Jawaharlal Nehru Engineering College	Bachelor's, Computer Science and Engineering	Aug 2020 - Jun 2024